Subject Name: Generative AI with Vector Databases
Subject Code: CSU 2217
Total Marks: 30
Exam: Term 2
Set F
B.Tech 2nd Year
Group - 1
Total Number = 31

Instructions:
The Exam is open book – You are allowed to bring books, lecture notes, slides.
No electronic gadgets are allowed.
Scientific Calculators are allowed for calculations.

Section A (5 * 2 = 10)

Q1. When we are working with the ef_search parameter of HNSW algorithm why recall becomes important parameter along with accuracy to be considered?

Q2. Two documents:

- ✓ One represented as sparse TF-IDF
- ✓ One as dense embedding

Which representation is more sensitive to noise and why

Q3. Two documents have cosine distance:

D1 = 0.1

D2 = 0.4

What does this imply geometrically? Why is cosine distance preferred over raw distance in text systems?

Q4. If M (max edges per node) is too small then what failure occurs in search quality?

Q5. When we are using BM25 vector base retriever explain what do you mean by k1 ~ 1.2 and b ~ 0.75?

Section B (10 * 2 = 20)

Q1.

a)

```python
def normalize(v):
    norm = sum(x*x for x in v) ** 0.5
    return [x / norm for x in v]
```

Why normalization changes ranking behavior?

b) You are given:
Vectors:
Q = (1, 2, 1, 0)
D1 = (2, 4, 2, 0)
D2 = (1, 1, 1, 1)
D3 = (0, 2, 1, 2)
Retriever Scores:

Doc	Semantic	Keyword
D1	0.85	0.6
D2	0.75	0.9
D3	0.65	0.7

Compute cosine similarity and rank documents.

Q2.

a)

```python
def search_all(query, vectors):
    best = None
    for v in vectors:
        if best is None or dist(v, query) < dist(best, query):
            best = v
    return best
```

Why is this unsuitable for large-scale systems?

b) Explain RRF_score(d) = Σ (1 / (rank_i(d) + k))